Shreya Ann Sebastian

Machine Learning, AI and Data Science | MSc Computer Science, TU Delft

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Dutch nationality | English (native), Dutch (professional working proficiency)

Profile

MSc Computer Science graduate from TU Delft (Artificial Intelligence track), with a BSc in Computer Science from Leiden (bioinformatics track). Built and evaluated machine learning models across unsupervised learning, deep learning, computer vision, natural language processing and predictive modelling. Designed, built and deployed Tech Jobs Radar, a public job board at techjobsradar.nl. Also worked with LLMs, including agentic AI, prompt engineering and LLM evaluation.

Education

MSc Computer Science 2023 - 2026
Delft University of Technology Artificial Intelligence Technology Track
Thesis: *Automated Discovery of Latent Activity Contexts from Multimodal Sensing for Peer-Interaction Analysis in Preschoolers with Hearing Loss*
Supervision: Socially Perceptive Computing Lab, TU Delft, with external supervision from the University of Miami

Unsupervised pipeline on UWB positioning and LENA audio data: a Gaussian Mixture Model found activity contexts, Dominant Sets clustering detected spatial groups, and per-context linear mixed models revealed inclusion effects absent from the pooled analysis. Redefined the research question when the planned data became unavailable and presented results to developmental psychologists.

BSc Computer Science 2020 - 2023
Leiden University Bioinformatics Track
Thesis: *Enhancing Pollen Classification Accuracy through the Use of Whole Stack Images in 3D Convolutional Neural Networks*
Supervision: Leiden Institute of Advanced Computer Science (LIACS), Leiden University

3D CNNs (custom ConvNet3D and ResNet3D) in PyTorch on microscope Z-stacks, with data augmentation, hyperparameter tuning and a whole-stack versus sub-stack comparison. Trained on about 9% of the data used in earlier 2D-projection work, a ResNet3D trained from scratch reached 96.7% test accuracy.

International Baccalaureate Diploma
International School Eindhoven
HL: Biology, Chemistry, Dutch B | SL: Environmental Systems & Societies, English A (Language & Literature), Mathematics

Projects

Tech Jobs Radar techjobsradar.nl
Personal project, deployed
Live job board that classifies tech roles and extracts structured fields from Dutch career sites. Designed, built and operated end to end on Kubernetes on AWS, with monitoring, alerting and automated evaluation tests in CI.
Tools: Python, FastAPI, Postgres, Kubernetes, AWS

RetroSTC-Net: Retroactive Consolidation for Neural Networks Repo
Personal research project, open source (MIT)
Retroactive weight consolidation inspired by synaptic tagging and capture, evaluated against continual learning baselines.
Tools: Python, PyTorch

LLM-Panel Annotation Framework Repo
Group research project, TU Delft
Three-LLM panel that annotates meeting transcripts against a codebook and measures inter-model agreement.
Tools: Python, OpenAI, Gemini and Groq APIs

Online Retail Purchase Prediction Repo

Personal project

Databricks pipeline on real retail transactions with k-means customer segmentation and propensity models, evaluated on four later months with a rolling time split. Boosted trees raised a category's average precision from 0.50 to 0.69.

Tools: PySpark, Databricks, MLflow, scikit-learn

Deep Learning Reproduction Project Repo

Group project, Deep Learning course, TU Delft

Reproduced a multi-view underwater image enhancement paper and wrote the technical blog post reporting the results.

Tools: Python, PyTorch

Fault-Tolerant Sensor Data Ingestion (Smart Home PoC) Repo

Software Architecture course, TU Delft

Sensor data ingestion validated under simulated network failures and high load, and tested in Docker through GitLab CI.

Tools: Python, RabbitMQ, Pydantic, Docker, GitLab CI

Personal Finance Engine Repo

Personal project

Finds subscriptions in bank statements; a local tool-calling LLM agent names hidden merchants with cited evidence.

Tools: Python, DuckDB, RapidFuzz, Ollama

Skills

Programming: Python, R, MATLAB, SQL, JavaScript, Java, C++, object-oriented programming, algorithms and data structures

Machine Learning: Supervised and unsupervised learning, classification, deep learning (CNNs), computer vision, natural language processing, reinforcement learning, model evaluation, explainability; PyTorch, TensorFlow, OpenCV, GPU training with CUDA

Generative AI: LLMs, agentic AI, prompt engineering, information extraction, LLM evaluation; OpenAI, Gemini and Groq APIs, Ollama

Data Science: Data analysis, probability and statistics, hypothesis testing, regression, feature engineering, clustering, customer segmentation, propensity modelling, experiment design, predictive modelling, data visualisation; pandas, NumPy, scikit-learn, matplotlib

Data Engineering: ETL and data pipelines, PySpark, Databricks (Delta Lake, Unity Catalog), SQLAlchemy, PostgreSQL, SQLite, DuckDB, Redis, RabbitMQ, web scraping

Software & MLOps: REST APIs (FastAPI), Pydantic, MLflow, Docker, Kubernetes (k3s, Helm), Terraform, AWS (EC2, S3, IAM), CI pipelines (GitHub Actions, GitLab CI), unit testing (pytest), monitoring (Prometheus), Git

Domains: Human-computer interaction, bioinformatics

Communication & Collaboration: Presenting, technical writing, Agile, Scrum

Extracurriculars

Basketball at DSBV Punch (TU Delft) and varsity level (International School Eindhoven), Student Council, Science Club